

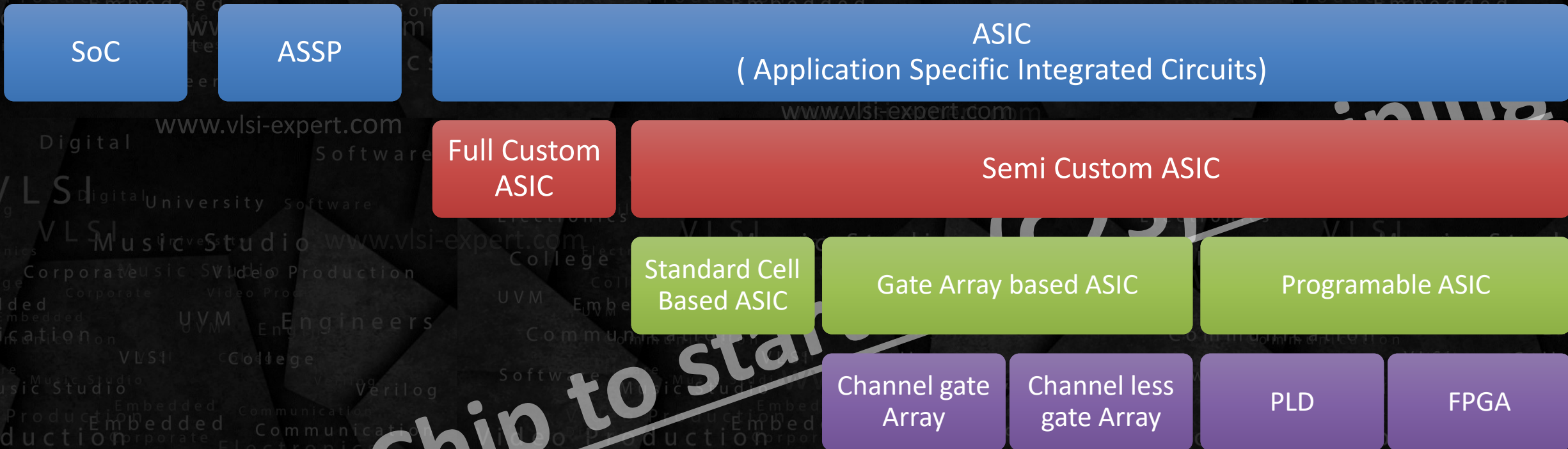
Topic: 1

VLSI Design Implementation Flow

Only for Chip to start-up (C2S) Training



Type of Design



SOC : System on Chip –

- An IC that integrates the major functional elements of a complete end-product into a single chip

ASSP: Application Specific Standard Parts –

- Chip designed for an application but for wider market – USB Controller

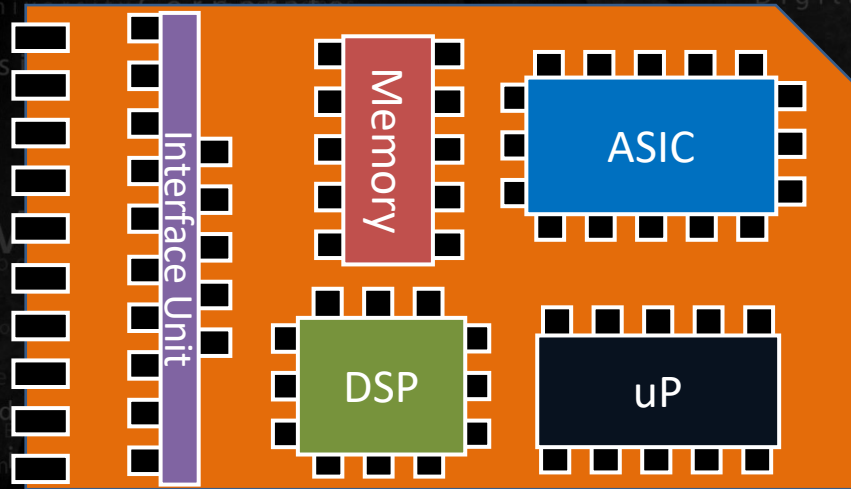
Full Custom: All logic cells and all mask layers customized

Semi Custom: All logic cells are pre-designed and some or all mask layers customized

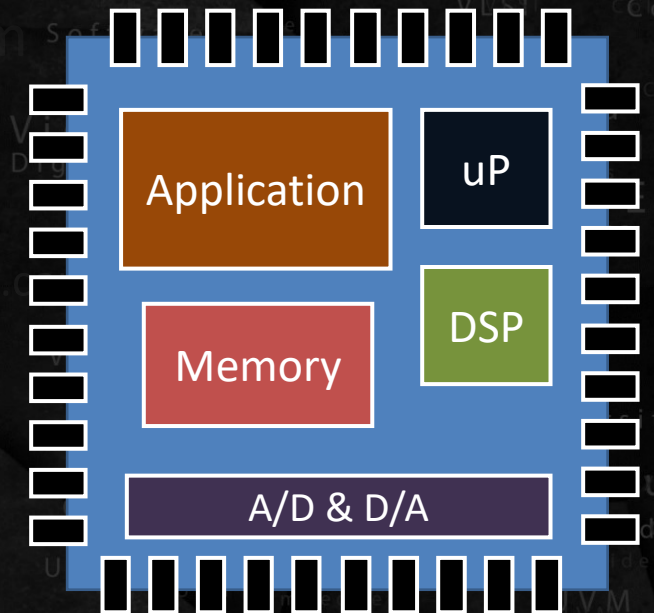
SOC (System on Chip)

SOC— System on VLSI Chip - that has all

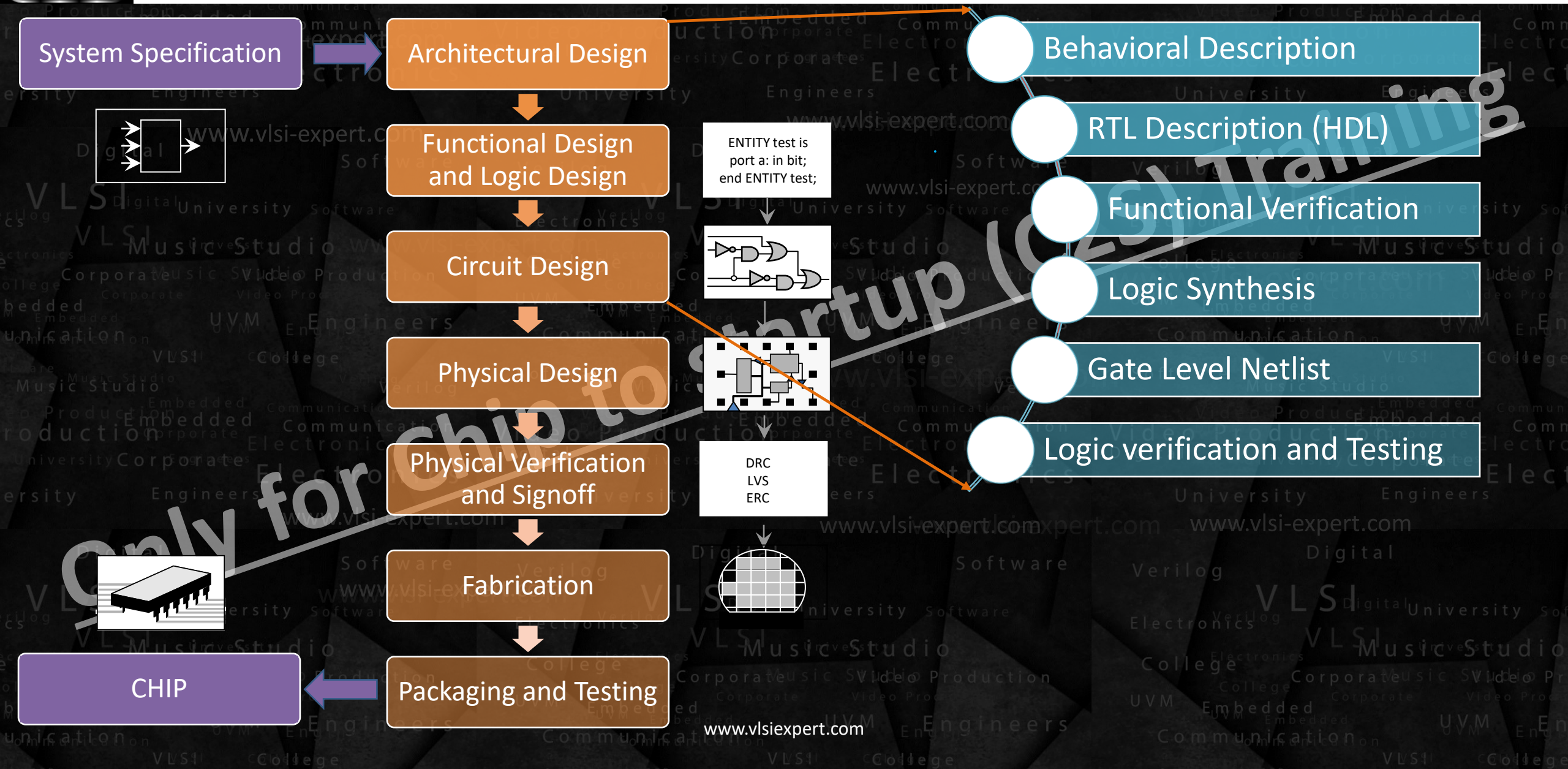
- Analog Circuits /Digital Circuits
- A/D and D/A
- Processors / Controllers
 - DSP /GPP (General purpose Processor, Microprocessor)
 - Micro-Controller (e.g ARM7, PIC18, others)
 - Other Processors (Media specific, IO Specific, network Related)
- ASICs (Application specific IPs)
 - Like USB port, Bluetooth, GPS interface, Wireless
- Memory (RAM/ROM)
- PLDs and FPGAs



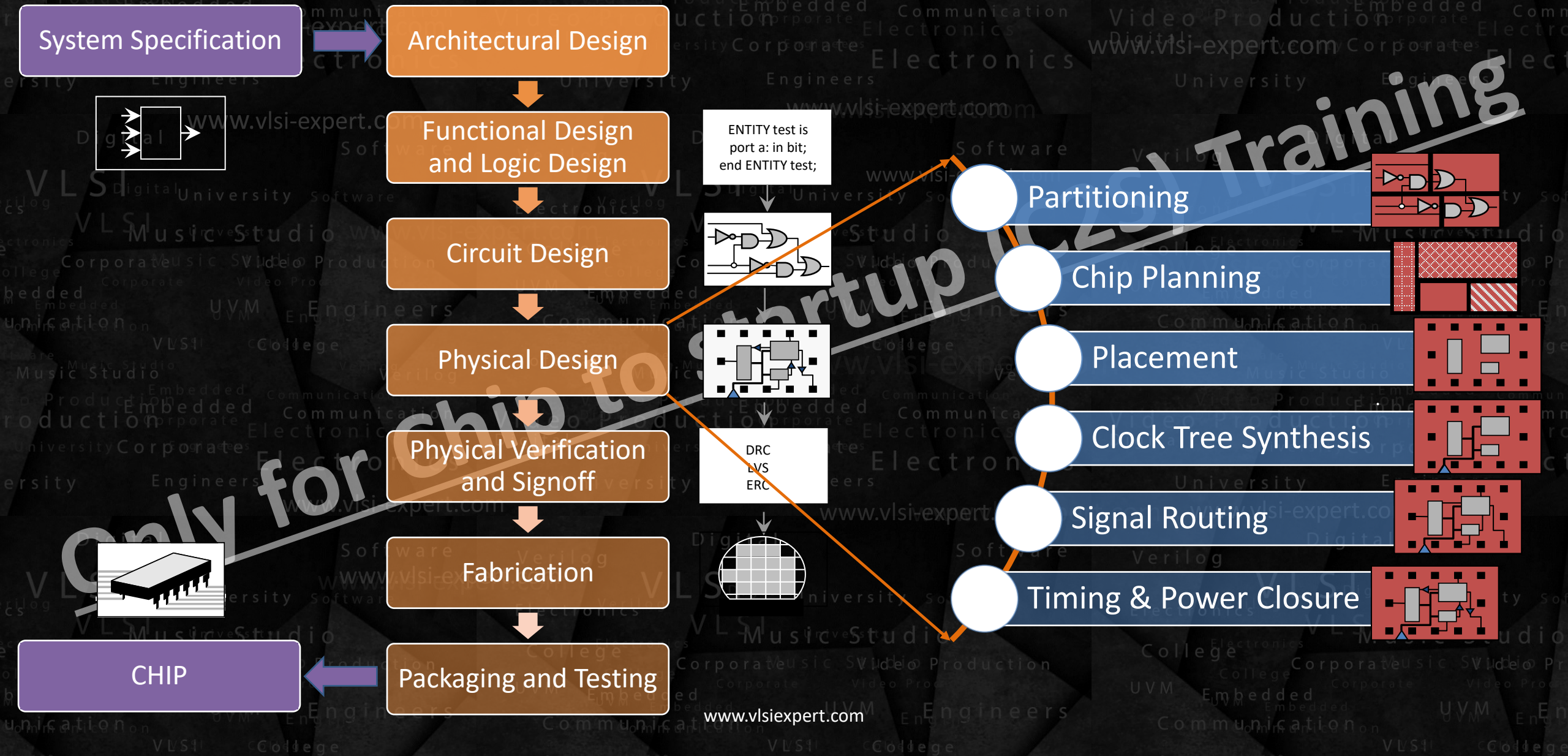
An IC that integrates the major functional elements of a complete end-product into a single chip
E.G : Single Chip Mobile Phone



VLSI Design Flow



VLSI Design Flow

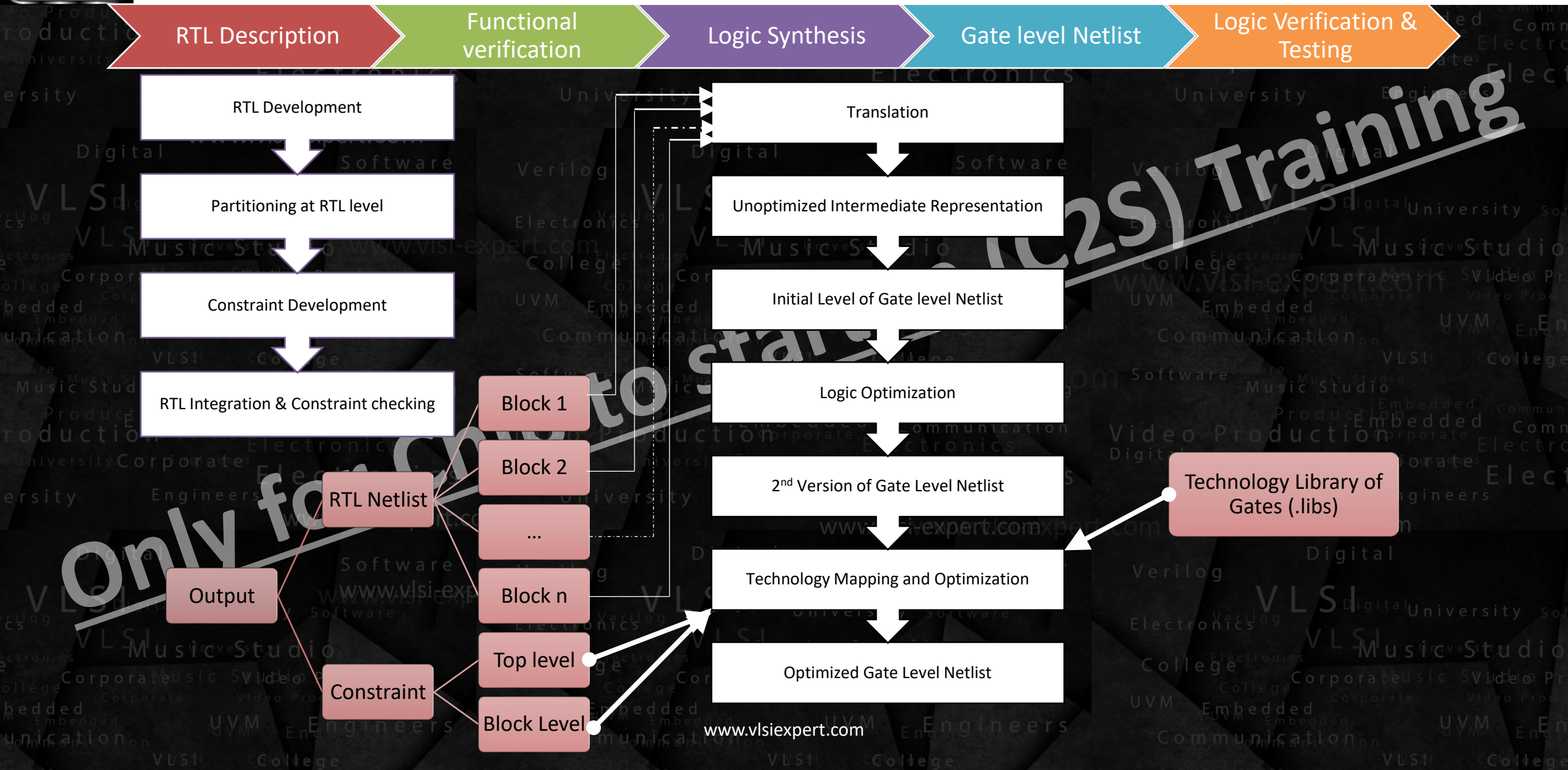


VLSI Design
Implementation Flow

RTL Designing & Integration

Only for Chip to start-up (C2S) Training

RTL to Gate Level Netlist

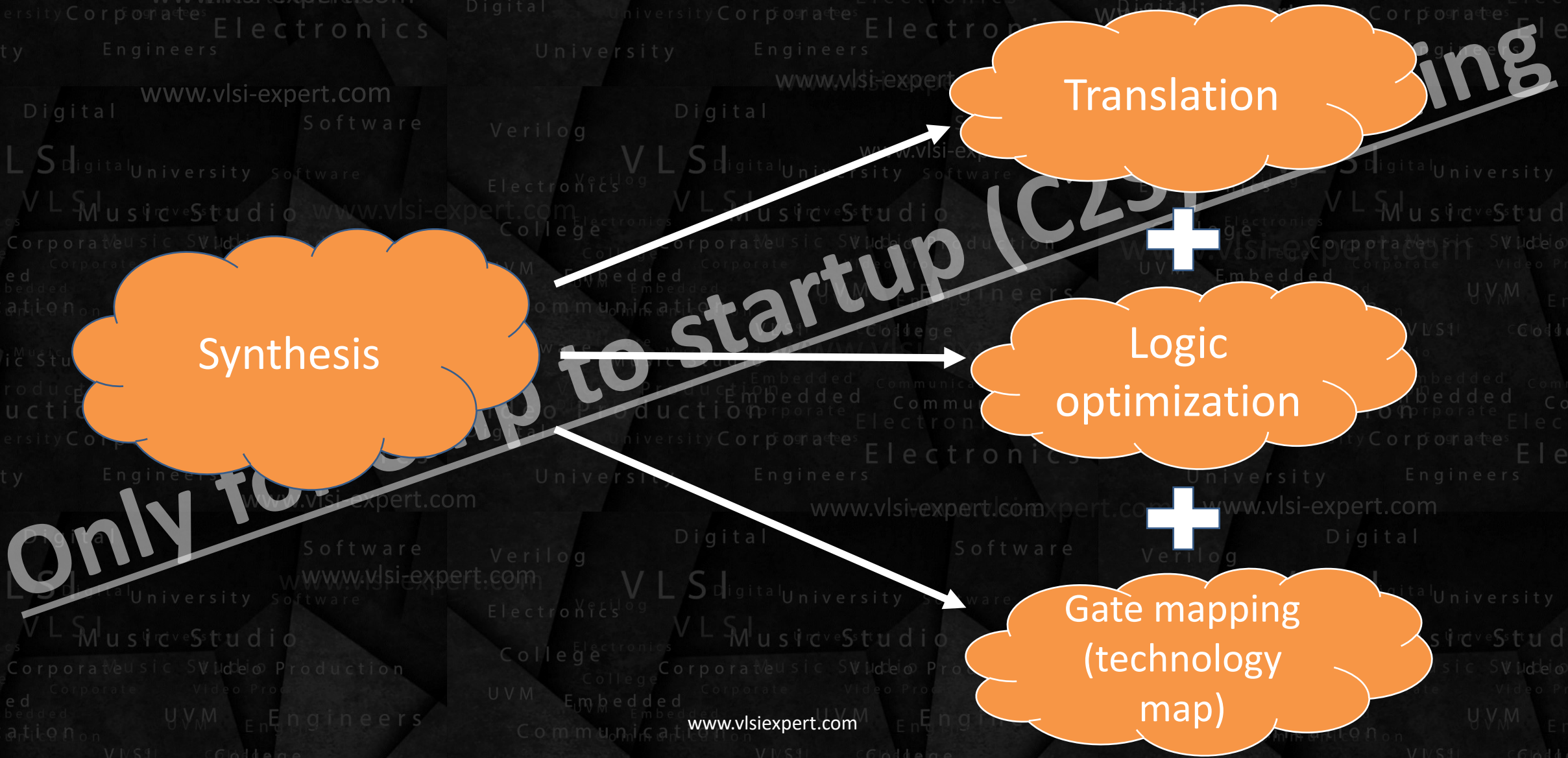


VLSI Design
Implementation Flow

Logic Synthesis & Optimization

Only for Chip to startup (C2S) Training

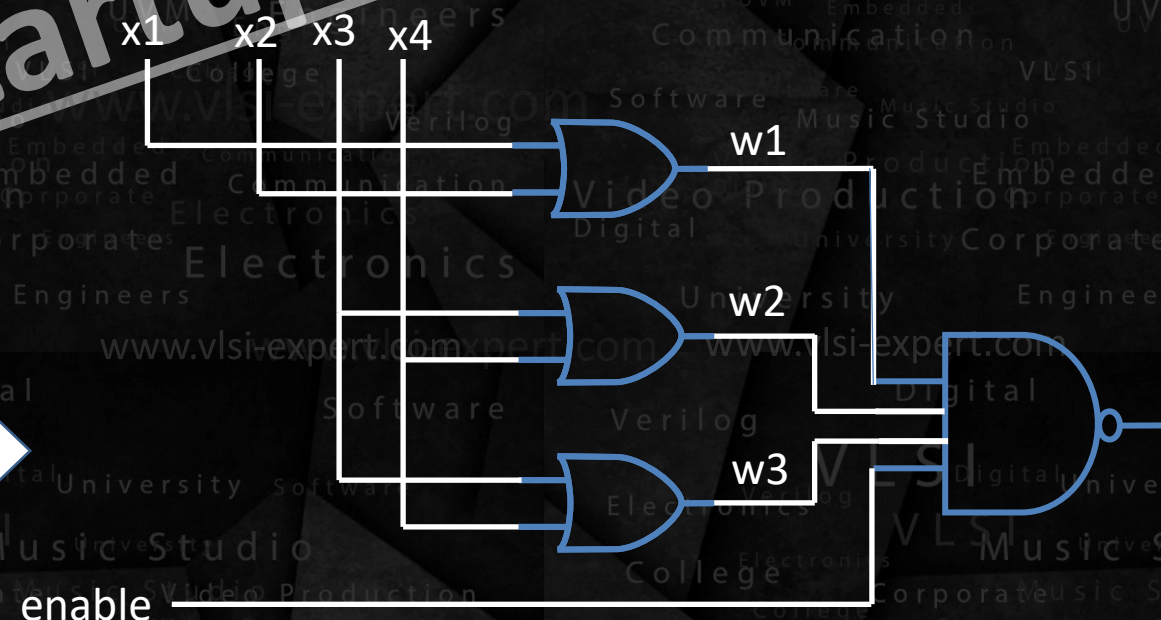
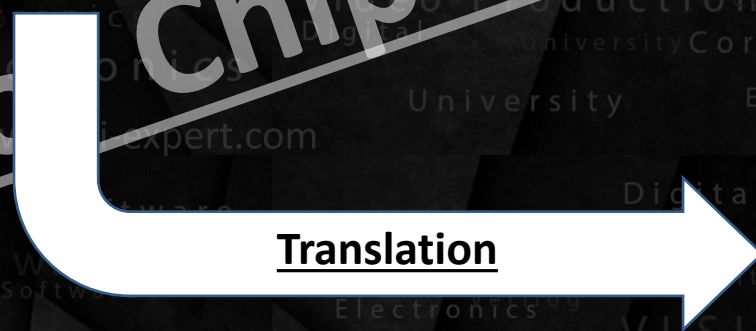
Synthesis Transformation



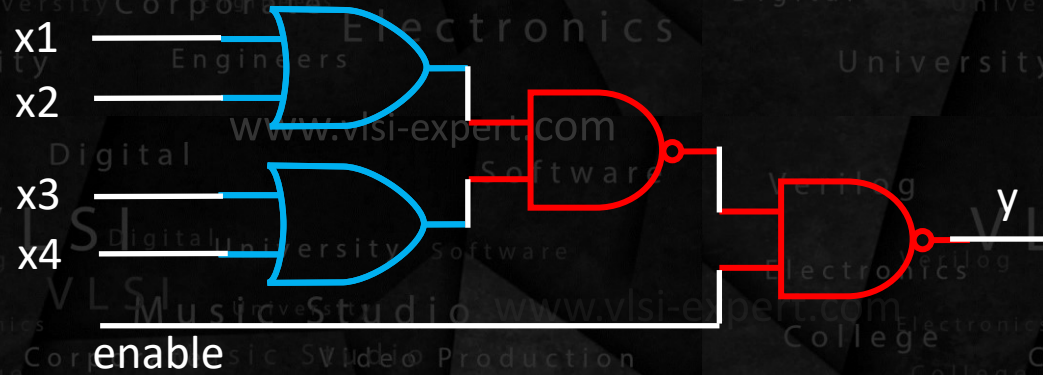
Logic Synthesis (Step 1)

```
module or_nand_1 (enable, x1, x2, x3, x4, y);
    input enable, x1, x2, x3, x4;
    output y;
    wire w1, w2, w3; or (w1, x1, x2);
    or (w2, x3, x4);
    or (w3, x3, x4); // redundant
    nand (y, w1, w2, w3, enable);
endmodule
```

Logic synthesis operates on Boolean equations and produces optimized combinational logic



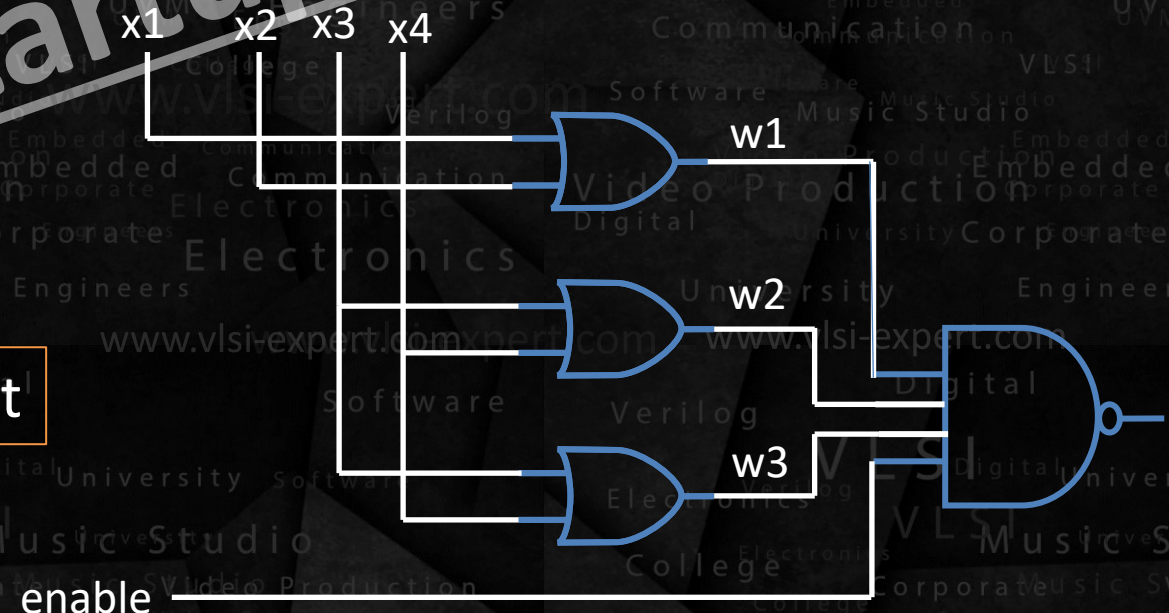
Logic Synthesis (Step 2)



Circuits after Synthesis

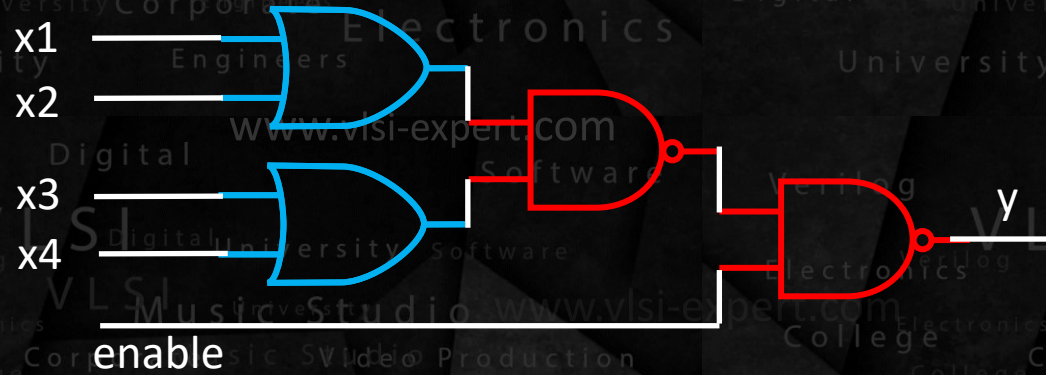
Logic Optimization

Pre- Optimized circuit



enable

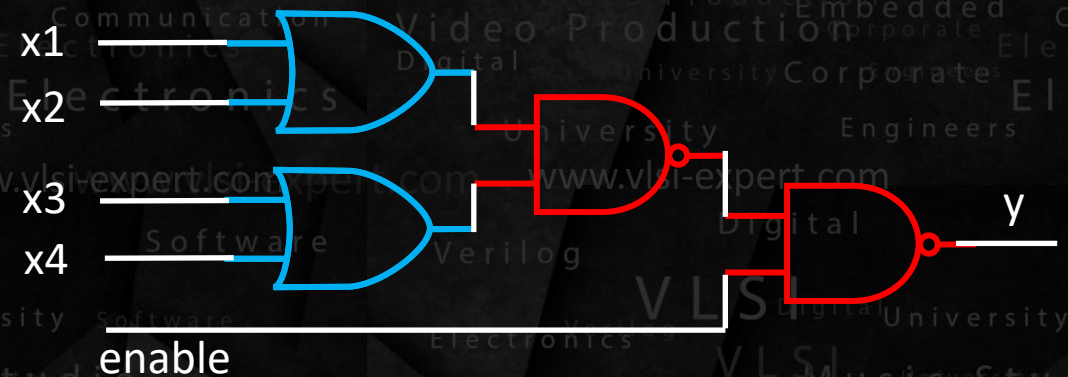
Logic Synthesis (Step 3)



Circuits with GPDK

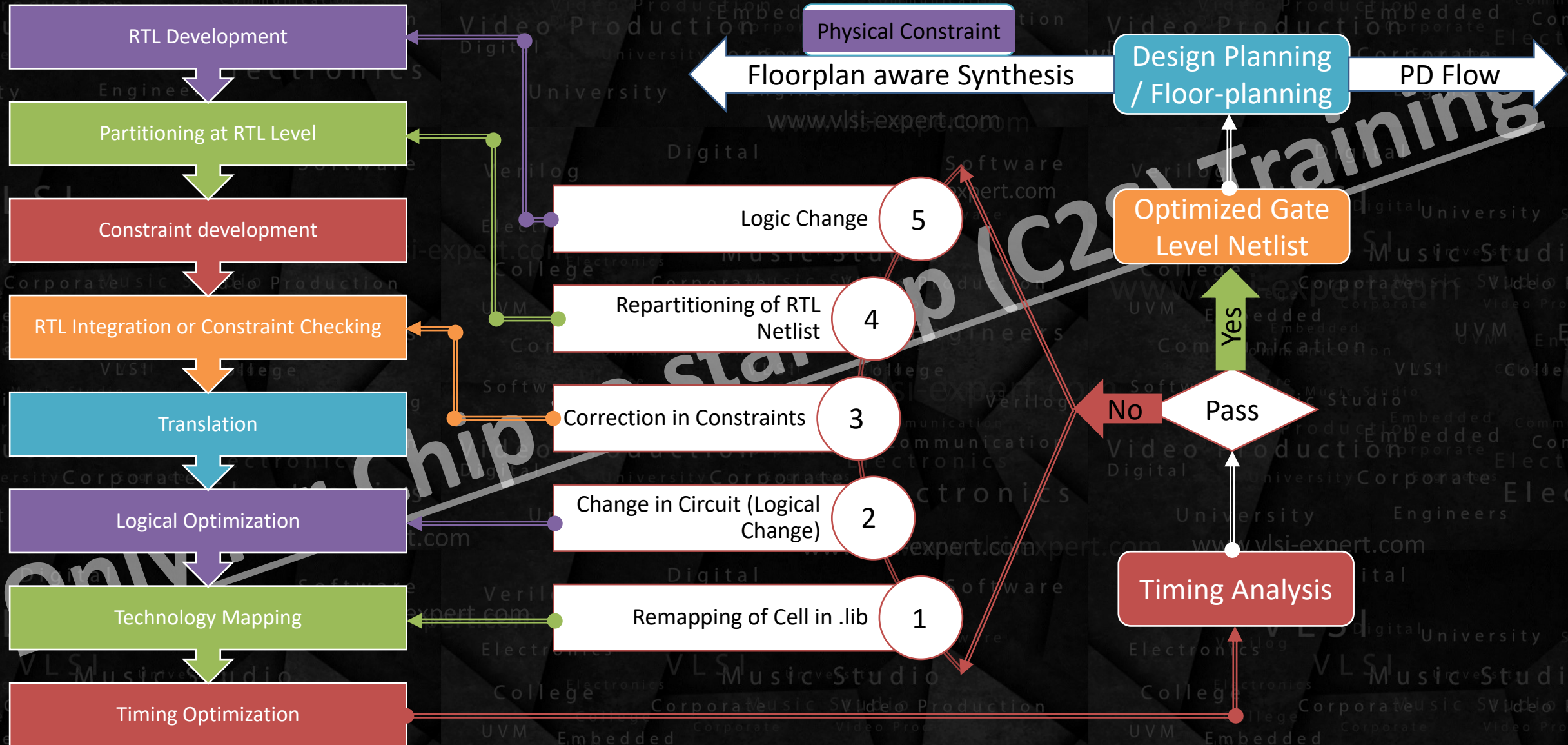
Circuits with TSMC 20LP PDK

Technology mapping



Synthesis Timing Optimization

Timing
Optimization



Floorplan aware Synthesis

